

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report reviews three primary patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile elastomer formulations. The target material is a butadiene/acrylonitrile copolymer with reduced acrylonitrile content, typically pursued to balance low-temperature flexibility, processability, and oil resistance while avoiding hydrogenated, carboxylated, epoxidized, or chlorinated variants unless directly relevant as comparative materials. The selected patents do not disclose a dedicated low-ACN NBR polymerization route in the narrow sense; instead, they address composition design and end-use performance of nitrile elastomers in demanding applications.

US9926439B2 discloses a carboxylated nitrile-butadiene rubber latex formulation based on blending two acrylonitrile terpolymers, with the blended acrylonitrile content stated as 17-45 wt% and the terpolymer ratio at 60:40 to 40:60. It emphasizes soft, natural-rubber-like force-strain behavior in dipped articles and provides a process context for latex compounding, ionic crosslinking, and sulfur vulcanization. US10385647B2 focuses on swellable packer compositions using a nitrile elastomer together with butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer; it is relevant because it explicitly includes nitrile rubber as a matrix option and gives practical high-temperature/high-pressure service conditions, though the patent is centered on swellable downhole seals rather than NBR synthesis. US10443339B2 is a broader elastomeric-material patent for bore tools and packers reinforced with carbon-based nanoplatelets; it mentions NBR among candidate elastomers but does not disclose NBR synthesis or composition control.

Across the landscape, the dominant trends are formulation-driven rather than monomer-feed-driven: blending of nitrile polymers to tune modulus, use of latex dipping and coagulant processes, sulfur/accelerator cure systems, and reinforcement or swell-control additives for downhole durability. For low-ACN NBR synthesis specifically, the evidence supports composition control by monomer ratio as the key lever, but the patents here provide limited direct polymerization recipes.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses swellable rubber compositions for downhole packers rather than a direct NBR polymerization process. The compositions combine a nitrile elastomer with butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer, and are processed into packers that swell in brine under heat and pressure. - Powder super absorbent polymer loading: about 125 Phr to about 175 Phr — Claim 2 - Powder super absorbent polymer loading: about 170 Phr to about 210 Phr — Claim 3 - Powder super absorbent polymer loading: at least about 150 Phr, about 175 Phr, about 200 Phr or about 225 Phr — Claim 4 - Powder super absorbent polymer fraction: at least about 50%, about 60%, about 70%, about 80%, about 90%, or about 95% by weight of total SAP — Claim 5 - Curing temperature: 330 degrees Fahrenheit — Example 1 - Curing pressure: 70 psi — Example 1 - Curing time: 2.5 hours — Example 1 - Test temperature: 275 degrees Fahrenheit — Example 1 - Test salinity: 16% brine — Example 1 - Test pressure requirement: 5000 psi — Example 1 - SAP loading: 110 Phr — Example 2 - SAP loading: 130 Phr — Example 2 - SAP loading: 150 Phr — Example 2 - SAP loading: 170 Phr — Example 2 - Test temperature: 200 degrees Fahrenheit — Example 2 - Test temperature: 275 degrees Fahrenheit — Example 2 - Test brine concentration: 5% brine by volume — Example 2 - Test brine concentration: 16% by volume brine — Example 2 - Predicted SAP loading for 75% swell: about 205 Phr — Example 3 - Predicted SAP loading for 75% swell: about 190 Phr — Example 3 - Maximum load for medium granular SAP: about 170 Phr — Example 3 - Predicted SAP loading for 75% swell: about 160 Phr — Example 3 - Maximum load for large granular SAP: about 150 Phr — Example 3 - Powder SAP loading: 205 Phr — Example 4 - NBR rubber loading: 10% Nipol DN3350 — Example 4 - HNBR loading: 10% Zetpol 2030L — Example 4 - Swelling time: 96 hour — Example 4 - Temperature: 200 degrees F. — Example 4 - Temperature: 275 degrees F. — Example 4 - Brine concentration: 5% brine — Example 4 - Brine concentration: 16% brine — Example 4 - Powder SAP loading: at least about 150 Phr, 175 Phr, 197 Phr, 200 Phr, 205 Phr or 225 Phr — Example 2A - Heat temperature: about 200 degrees Fahrenheit — Example 3A - Heat temperature: about 275 degrees Fahrenheit — Example 4A - Brine concentration: about 5% weight/volume — Example 5A - Brine concentration: about 16% weight/volume — Example 6A - Swelling: at least about 50%, 60%, 65%, 70%, 75%, 80%, or 90% by volume — Example 7A - Duration of time: about 2, 3, 4, or 5 days — Example 8A - Pressure resistance: at least about 3000, 4000, 5000, 6000, 70000, or 8000 psi — Example 9A - Pressure resistance: at least about 5000 psi — Example 11A - Temperature resistance: at least about 200 degrees Fahrenheit — Example 11A - Temperature resistance: at least about 275 degrees Fahrenheit — Example 12A

Relevant Experimental Evidence - Example 1: A packer using Composition B was cured at 330°F and 70 psi for 2.5 hours and then withstood 16% brine at 275°F and about 300 psi, later surviving >5000 psi differential pressure. - Example 2: Powder, medium granular, and large granular SAPs were compared at 110-170 Phr, showing strong dependence of swelling and leaching on SAP type and loading. - Example 3: The data were used to predict 75% swell targets, with powder SAP supporting the highest practical loading and best manufacturability. - Example 4: A 205 Phr powder SAP formulation with 10% NBR or 10% HNBR showed excellent green strength and acceptable processing, with comparable swelling performance.

Technical Relevance This patent is relevant because it explicitly uses nitrile elastomer as a matrix component in a swellable rubber system and provides practical high-temperature/high-pressure service conditions. While not a synthesis patent for low-ACN NBR, it shows how nitrile rubber formulations are tuned for downhole sealing performance.

Patent 2

- Patent Number: US9926439B2
- Patent Title: Nitrile rubber article having natural rubber characteristics
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a carboxylated nitrile-butadiene rubber latex formulation made by blending a first and second acrylonitrile terpolymer in a defined ratio. The latex is designed for dipped elastomeric articles with soft, natural-rubber-like force-strain behavior, using ionic crosslinking and sulfur vulcanization to balance softness, strength, and processability. - Acrylonitrile content of blended terpolymers: between 17 and 45% by weight — Abstract/Claim 1 - Methacrylic acid content: less than 15% by weight — Abstract/Claim 1 - First/second terpolymer ratio: 60:40 to 40:60 — Claim 1 - Alkali agent: 0-1.5 parts per 100 dry parts of rubber — Claim 3 - Metallic oxide: 0.5-1.5 parts per 100 dry parts of rubber — Claim 3 - Sulfur: 0.5-1.5 parts per 100 dry parts of rubber — Claim 4 - Rubber accelerator: 0.5-1.5 parts per 100 dry parts of rubber — Claim 4 - Formulation pH: 9 to 12.5 — Claim 5 - Solids content of nitrile emulsion: 40-45% — Example section - Reduced solids content: approximately 23% — Example section - Further reduced solids content: approximately 20% — Example section - Mould preheat temperature: approximately 55-60 degrees C. — Example section - Preferred mould preheat temperature: about 58 degrees C. — Example section - Reheat temperature after coagulant dip: approximately 70 degrees C. $\pm$ 5 degrees C. — Example section - Drying temperature: about 80 degrees C. to below about 100 degrees C. — Example section - Stress relaxation time: about 10-15 minutes — synthesis/background section - Elastic modulus: about 3 MPa to about 6 MPa — summary section - Tensile strength: about 30 or 32 MPa to about 56 or 58 MPa — summary section - Glove thickness: about 0.05 mm to about 0.10 or 0.11 mm —

summary section - Conventional glove thickness: about 0.14 ± 0.02 mm — summary section

Relevant Experimental Evidence - Example 1: Demonstrated low force at 400% elongation for the inventive nitrile article relative to comparative nitrile gloves. - Example 2: Showed a similar soft-stretch response with the inventive formulation. - Example 3: Confirmed the inventive article maintained low force at high elongation while preserving nitrile properties. - Example 4: Reinforced the consistency of the soft-nitrile behavior across examples. - Example 12: Comparative data showed the inventive examples required only a fraction of the force of eight comparative nitrile examples.

Technical Relevance This is the most directly relevant patent for low-ACN NBR because it explicitly defines acrylonitrile content in the 17-45 wt% range and uses blending of nitrile terpolymers to tune softness and film formation. It provides a clear composition-control strategy for nitrile rubber articles, even though the focus is on carboxylated latex rather than a pure low-ACN NBR synthesis route.

Patent 3

- Patent Number: US10443339B2
- Patent Title: Elastomeric materials
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses elastomeric bore-tool and packer components reinforced with carbon-based nanoplatelets in an elastomeric matrix. It is a materials and device patent rather than a nitrile polymerization patent, and it broadly contemplates elastomers such as NBR, HNBR, AFLAS, and EPDM as candidate matrices. - Carbon-based nanoplatelets content: less than approximately 10 percent by weight — Claim 1 - HPHT environment pressure: up to about 138 MPa — Detailed description - HPHT environment temperature: up to about 205 degrees C. — Detailed description - Ultra-HPHT environment pressure: up to about 241 MPa — Detailed description - Ultra-HPHT environment temperature: up to about 260 degrees C. — Detailed description - HPHT-hc environment pressure: greater than about 241 MPa — Detailed description - HPHT-hc environment temperature: greater than about 260 degrees C. — Detailed description - Young's modulus: about 3 MPa — Detailed description - HNBR mechanical property decrease temperature: about 160 degrees C. — Detailed description - AFLAS mechanical property decrease temperature: about 220 degrees C. — Detailed description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it mentions NBR as one possible elastomer matrix for high-performance bore tools. It does not teach NBR synthesis or acrylonitrile-content control, but it does show the service environment and reinforcement concepts where NBR-based materials may be deployed.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US9926439B2 is the only patent in the set that explicitly controls acrylonitrile content; it states 17-45 wt% ACN and a 60:40 to 40:60 blend ratio of two acrylonitrile terpolymers.
- US10385647B2 uses nitrile elastomer as a formulation component in swellable packers, but the patent emphasis is on SAP loading, brine swelling, and downhole durability rather than monomer composition.
- US10443339B2 is a reinforcement/materials patent for bore tools and packers; it mentions NBR as a possible elastomer but provides no NBR composition or polymerization recipe.
- Process-wise, US9926439B2 relies on latex blending, coagulant dipping, ionic crosslinking, and sulfur vulcanization; US10385647B2 relies on cured swellable rubber formulations with powder SAP; US10443339B2 relies on elastomeric matrices reinforced with carbon-based nanoplatelets.
- Temperature and pressure service conditions are most explicit in US10385647B2 and US10443339B2, with US10385647B2 showing 200-275°F brine testing and >5000 psi resistance, and US10443339B2 citing HPHT conditions up to 138 MPa and 205°C.
- For low-ACN NBR synthesis, the strongest evidence is composition tuning in US9926439B2; the other two patents are application-oriented and only indirectly informative.

4. CONCLUSION

The evidence base shows that low-acrylonitrile NBR development is driven primarily by composition control rather than by novel monomer-polymerization chemistry in these patents. US9926439B2 provides the clearest compositional guidance, with blended acrylonitrile terpolymers at 17-45 wt% ACN and a 60:40 to 40:60 blend ratio, supporting soft, natural-rubber-like behavior while retaining nitrile performance. US10385647B2 and US10443339B2 are valuable application patents for downhole elastomer service, but they do not materially advance low-ACN NBR synthesis itself.

Based on the disclosed evidence, the most defensible synthesis/design direction for low-ACN NBR is to target the lower end of the nitrile-content window while maintaining latex processability and cure balance, then validate performance in brine, heat, and pressure. For practical formulation work, the data support using latex-based processing, controlled ionic crosslinking, and sulfur cure systems, with downstream reinforcement or SAP additions only as needed for the final application. The patents collectively suggest that low-ACN NBR should be optimized by monomer ratio and

formulation architecture, not by aggressive post-modification.

5. REFERENCES

- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>
- US9926439B2 | Nitrile rubber article having natural rubber characteristics | Not disclosed | US | 2018 | <https://patents.google.com/patent/US9926439B2/en>
- US10443339B2 | Elastomeric materials | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10443339B2/en>